An explicit sufficient width for mean-field Bayesian circuit learning

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Abstract

For fixed-depth tanh networks with mean-field Gaussian scaling, a global Boolean margin restriction, and exact noiseless Bayesian conditioning, we quantify the width threshold in a polynomial-sample learning theorem. The proof replaces an asymptotic large-deviation comparison by a finite-width method-of-types bound. Gaussian innovations are partitioned into finitely many cells, and each empirical type is lifted to a finite-entropy Gaussian conditional law. Entropy controls both the lift's second moment and its tail mass; these controls propagate through the hidden covariances. The resulting threshold is explicit and depends on the target only through its bounded-weight network complexity. It is double exponential in input dimension, while the sample bound above the threshold remains polynomial in input dimension and target complexity. A polynomial threshold, and a polynomial sample bound valid at every width above target complexity, remain open for deep networks. We include the variational and compression ingredients so that the result can be read independently.

1 Model and result

Let $d \geq 1$, $L \geq 2$, $m = 2^d$, and $r = L - 1$. The input space is $\mathcal{X} = \{-1, 1\}^d$, with augmented input $\bar{x} = (1, x)$. Every hidden layer has width N, and

$$h^0(x) = \bar{x}, \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no additional biases. The independent centered Gaussian weights have standard deviations $d^{-1/2}$ in the first layer, $N^{-1/2}$ in internal layers, and N^{-1} in the readout. Their law is μ_N. For a Boolean function g write

$$A_{N,g} = \{w : g(x) z_w(x) > a \text{ for every } x\}, \qquad a = 1/2, \qquad A_N = \bigcup_g A_{N,g}.$$

The function prior is $\pi_N(g) = \mu_N(A_{N,g})/\mu_N(A_N)$. Learning conditions this prior on agreement with independent noiseless examples $(x_i, f(x_i))$, with $x_i \sim D$. We denote the posterior by $\Pi_N(\cdot \mid S_n)$ and its average error by

$$R_D(\Pi_N) = \mathbb{E}_{g \sim \Pi_N} \mathbb{P}_{x \sim D}\{g(x) \neq f(x)\}.$$

Let $C = C_L(f)$ be the least integer for which f has a depth-L, width-C implementation with every weight bounded by C and strict global margin $1/2$.

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Our threshold uses the following explicit target-cost bound. Set

$$\begin{aligned} b_L &= 2+\sqrt{2\log(2L)}, & A_* &= 2+3C^2+b_L,\\ \rho &= [2+9C^2(d+L+1)A_*^{L-2}]^{-1},\\ B_0 &= d(d+1)C^2+2(L-2)C^4+36C^4,\\ A &= 2B_0+4[d+2+(L-2)C]/\rho+2, & K &= 4A+4. \end{aligned}$$

The cylinder construction in Appendix A proves, for all $N \geq C$,

$$\mu_N(A_{N,f}) \geq e^{-AN/2}. \tag{1}$$

All these quantities are polynomially bounded in d, C at fixed L.

Define the quantization parameters

$$\begin{aligned} B &= 4K+2m\log 2, & b &= 2\sqrt{mB}, & \beta &= \frac{1}{128mK},\\ c &= \left(\frac{\beta}{4b^2}\right)^{2^{r-1}}, & t &= \frac{c}{8}, & \Delta &= \frac{c}{8\sqrt{2m}},\\ R^2 &= 4\left[\frac{K+\log 2}{t}+\frac{m}{2}\log 2\right], & q &= 1+\left[1+\frac{2R\sqrt{m}}{\Delta}\right]^m. \end{aligned} \tag{2}$$

Here q is an upper bound on the number of quantization cells; it is an integer, and should not be confused with a prediction-confidence threshold. For $\varepsilon \in (0,1)$ set

$$N_0 = \left\lceil \max\left\{C, 1, \left(\frac{8rq}{K}\right)^2, \frac{4m\log 2}{K}, \frac{8}{K}\log\frac{6}{\varepsilon}\right\}\right\rceil. \tag{3}$$

Theorem 1.1 (Explicit sufficient width)**.** *At every fixed depth L, there is a polynomial p_L such that the following holds for every target of finite complexity $C = C_L(f)$, every $\varepsilon, \delta \in (0,1)$, every input distribution D, and every $N \geq N_0$ from (3). If*

$$n \geq \frac{6}{\varepsilon}\left[p_L(d,C)+\log\frac{1}{\delta}\right],$$

then, with probability at least $1-\delta$ over the training sample,

$$R_D(\Pi_N) \leq \frac{\varepsilon}{3}, \qquad \mathbb{P}_{x\sim D}\left\{\Pi_N(g(x)=f(x)\mid S_n) > \frac{2}{3}\right\} \geq 1-\varepsilon.$$

An explicit substitute for $p_L(d,C)$ is $B_L(d,2K)$ from the finite-network count in Appendix E. The threshold satisfies

$$N_0 \leq C+\exp\Big[O_L(2^d(d+\log K))\Big]+O\left(K^{-1}\log\frac{1}{\varepsilon}\right).$$

In particular, the width threshold is quantified, but is not polynomial in input dimension. The theorem does not establish the same polynomial sample bound for every width $N \geq C$.

The sample bound and the width threshold play distinct roles. The sample bound is polynomial in the target complexity and independent of N. The threshold is much larger: quantizing all 2^d coordinates of a Gaussian field leads to an exponential number of cells in 2^d. For one hidden layer, an existing direct norm argument already gives a polynomial sample bound for every $N \geq C$; the present threshold is useful as a general-depth quantitative statement.

2 An exact representation and a variational complexity

Enumerate the inputs by $1, \dots, m$ and let $\Gamma_m = \mathcal{N}(0, I_m)$. For a positive semidefinite matrix Q, put

$$T_Q(u) = \tanh(Q^{1/2}u), \qquad F_Q(u) = T_Q(u)T_Q(u)^{\mathsf{T}}, \qquad Q_0(x,y) = \bar{x} \cdot \bar{y}/d.$$

Given probability laws $\eta_1, \dots, \eta_r$ on $\mathbb{R}^m$ and $v \in \mathbb{R}^m$, recursively set

$$Q_s = \int F_{Q_{s-1}}(u)\, d\eta_s(u), \qquad z = Q_r^{1/2} v.$$

Define

$$I(z) = \inf\left\{ \sum_{s=1}^{r} D(\eta_s \| \Gamma_m) + \frac{1}{2}\|v\|_2^2 : (\eta_1, \dots, \eta_r, v) \text{ gives } z \right\}, \qquad J(g) = \inf_{g_x z_x \geq a \ \forall x} I(z). \tag{4}$$

Entropy contraction makes this equal to the hidden-feature formula proved in Appendix B. We use that appendix and the quadratic-tilt compression in subsequent appendices only to obtain the polynomial count of J sublevels. The new finite-width probability estimate below does not invoke an unquantified large-deviation limit.

For each layer s let $U_{s,1}, \dots, U_{s,N}$ be independent standard Gaussian vectors, independently across layers, and let $\eta_{s,N} = N^{-1}\sum_i \delta_{U_{s,i}}$. Independently take $V_N \sim \mathcal{N}(0, I_m/N)$. The recursion above gives empirical covariances $Q_{s,N}$. Exactly for every finite N,

$$Z_N \overset{\text{law}}{=} Q_{r,N}^{1/2} V_N. \tag{5}$$

Indeed, conditional on the preceding layer, the next preactivation rows are independent Gaussian fields with covariance equal to that layer's empirical activation covariance. The readout has conditional covariance $Q_{r,N}/N$. This proves the representation, including singular covariance matrices.

3 A finite-entropy lift of an empirical type

Partition the Euclidean ball $\{u : \|u\|_2 \leq R\}$ into intersections with axis-aligned grid cells of side $\Delta/\sqrt{m}$, and add its complement as one outside cell. Ignore cells of zero Gaussian probability; the sampled innovations avoid their union almost surely. Every inside cell has diameter at most Δ, and their total number, including the outside cell, is at most q in (2). Write the Gaussian cell probabilities as α_j.

Lemma 3.1 (Entropy controls the lift)**.** *For a cell-frequency vector $p = (p_j)$ define*

$$\eta^p = \sum_j p_j\, \Gamma_m(\,\cdot \mid \textit{cell } j).$$

Then $D(\eta^p \| \Gamma_m) = D(p\|\alpha)$. If this entropy is at most K, then

$$\int \|u\|_2^2\, d\eta^p(u) \leq B, \qquad p_{\text{out}} \leq t.$$

Proof. The lift has density p_j/α_j on cell j, proving the entropy identity. The entropy inequality at exponent $1/4$ gives

$$\frac{1}{4}\int \|u\|_2^2\, d\eta^p \leq D(\eta^p\|\Gamma_m) + \log \mathbb{E}_{\Gamma_m} e^{\|U\|_2^2/4} \leq K + \frac{m}{2}\log 2.$$

For an initially unbounded test function this follows by truncation and monotone convergence. It is the stated moment bound. Gaussian Markov gives

$$\alpha_{\mathrm{out}} \le 2^{m/2} e^{-R^2/4} \le \exp[-(K + \log 2)/t].$$

Data processing to the two-cell inside/outside partition and the bound on binary entropy imply

$$D(p\|\alpha) \ge p_{\mathrm{out}} \log(1/\alpha_{\mathrm{out}}) - \log 2.$$

It follows that $p_{\mathrm{out}} \le t$. □

Lemma 3.2 (Uniform covariance comparison)**.** *Suppose an empirical measure η has cell type p, with $D(p\|\alpha) \le K$. For every positive semidefinite Q with diagonal entries at most two,*

$$\left\| \int F_Q \, d\eta - \int F_Q \, d\eta^p \right\|_{\max} \le c.$$

For any two positive semidefinite matrices Q, Q' and any lift of entropy at most K,

$$\left\| \int F_Q \, d\eta^p - \int F_{Q'} \, d\eta^p \right\|_{\max} \le b\sqrt{\|Q - Q'\|_{\max}}.$$

Proof. Each entry of F_Q is bounded in absolute value by one and is $2\|Q^{1/2}\|_{\mathrm{op}}$-Lipschitz. If the diagonal of Q is bounded by two, then $\|Q\|_{\mathrm{op}} \le \operatorname{tr} Q \le 2m$. Couple each empirical point with an independent conditional Gaussian draw in its own cell. Inside cells the difference is at most $2\sqrt{2m}\Delta$; outside it is at most two. Lemma 3.1 therefore bounds the integrated difference by

$$2\sqrt{2m}\Delta + 2t = c/2 \le c.$$

For the second claim, the positive square-root inequality gives

$$\|Q^{1/2} - (Q')^{1/2}\|_{\mathrm{op}} \le \|Q - Q'\|_{\mathrm{op}}^{1/2} \le \sqrt{m\|Q - Q'\|_{\max}}.$$

The entrywise product bound and Cauchy–Schwarz, with Lemma 3.1, give an upper bound of $2\sqrt{m\|Q - Q'\|_{\max}}\sqrt{B}$, as required. □

4 A nonasymptotic rate-sublevel bound

Proposition 4.1 (Low type cost implies low variational complexity)**.** *Let p_s be the types of any empirical innovation measures $\eta_{s,N}$, and let $v \in \mathbb{R}^m$. If*

$$H := \sum_{s=1}^{r} D(p_s\|\alpha) + \frac{1}{2}\|v\|_2^2 \le K,$$

and $Q_{r,N}^{1/2} v$ has strict margin a and sign g, then $J(g) \le 16K/9 < 2K$.

Proof. Replace each empirical law by its conditional Gaussian lift η_s^* and let Q_s^* be the resulting recursive covariances. Both recursions start at Q_0, whose diagonal is at most two; every later covariance has diagonal at most one. For $e_s = \|Q_{s,N} - Q_s^*\|_{\max}$, Lemma 3.2 gives

$$e_0 = 0, \qquad e_s \le c + b\sqrt{e_{s-1}}.$$

Since $b \geq 1$ and $c \leq 1$, induction yields

$$e_s \leq 4b^2 c^{1/2^{s-1}}, \qquad e_r \leq \beta.$$

For the induction step, $b\sqrt{4b^2 c^{1/2^{s-2}}} = 2b^2 c^{1/2^{s-1}}$, and the additional c is bounded by the same power of c. The lifted output $z^* = (Q_r^*)^{1/2} v$ satisfies

$$\|z^* - Q_{r,N}^{1/2} v\|_\infty \leq \sqrt{m\beta}\|v\|_2 \leq \sqrt{2mK\beta} = a/4.$$

Thus $(4/3)z^*$ has strict margin a and sign g. It is obtained from the lifted laws and output vector $(4/3)v$. Its cost in (4) is at most

$$\sum_s D(p_s\|\alpha) + \frac{16}{9}\frac{1}{2}\|v\|_2^2 \leq \frac{16}{9}H \leq \frac{16}{9}K.$$

This proves the claim. □

Theorem 4.2 (Finite-width aggregate bound)**.** *Let $\mathcal{A}_K = \{g : J(g) \leq 2K\}$. At every width $N \geq 1$,*

$$\mu_N\left(\bigcup_{g \notin \mathcal{A}_K} A_{N,g}\right) \leq (N+1)^{rq}\, 2^{m/2} e^{-NK/2}. \tag{6}$$

Proof. The probability of a fixed empirical type p from N independent draws with cell probabilities α is at most $e^{-ND(p\|\alpha)}$. Indeed, the multinomial coefficient is at most $e^{NH(p)}$. There are at most $(N+1)^q$ possible types per layer. For a fixed tuple of types put $D_* = \sum_s D(p_s\|\alpha)$. Independence across layers and from V_N gives

$$\begin{aligned}
&\mathbb{P}\Big\{\text{these types, } D_* + \tfrac{1}{2}\|V_N\|_2^2 > K\Big\} \\
&\quad \leq e^{-ND_*} \exp\{ND_*/2 - NK/2\}\mathbb{E}\exp\{N\|V_N\|_2^2/4\} \\
&\quad = 2^{m/2}\exp\{-ND_*/2 - NK/2\} \leq 2^{m/2} e^{-NK/2}.
\end{aligned}$$

The exponential bound remains valid when $D_* > K$. By Proposition 4.1, every full-margin draw whose sign function lies outside $\mathcal{A}_K$ has total cost greater than K. Summing over at most $(N+1)^{rq}$ tuples proves (6). □

This controls the entire union of undesired functions. It does not sum an asymptotic estimate separately over all 2^{2^d} Boolean functions. The large dimensional dependence enters instead through quantizing Gaussian innovations in $\mathbb{R}^{2^d}$.

5 Proof of the learning theorem and the remaining goal

Proof of Theorem 1.1. The target-mass bound, Theorem 4.2, and cancellation of the global-margin normalization give

$$\frac{\pi_N(\mathcal{A}_K^c)}{\pi_N(f)} \leq \exp\left[rq\log(N+1) + \frac{m}{2}\log 2 - \frac{N}{2}(K - A)\right].$$

For $N \geq 1$, $\log(N+1) \leq \sqrt{N}$. The first two nonconstant constraints in (3) therefore give

$$rq\log(N+1) \leq NK/8, \qquad (m/2)\log 2 \leq NK/8.$$

Since $A < K/4$, the ratio is at most $e^{-NK/8} \le \varepsilon/6$. In particular $f \in \mathcal{A}_K$: otherwise the ratio would be at least one. Appendix E gives $\log|\mathcal{A}_K| \le B_L(d, 2K)$, a polynomially bounded quantity.

Every particular function with D-error greater than $\varepsilon/6$ survives n noiseless examples with probability at most $e^{-n\varepsilon/6}$. A union bound over $\mathcal{A}_K$ shows that, except on a sample event of probability at most δ, every surviving function in this set has error at most $\varepsilon/6$ for the stated n. For every sample the posterior denominator is at least $\pi_N(f)$; therefore its mass outside $\mathcal{A}_K$ is at most $\varepsilon/6$. The total posterior error is at most $\varepsilon/3$. On each input where the correct-label probability is at most $2/3$, the posterior error probability is at least $1/3$. The mass of these inputs is at most $3R_D(\Pi_N) \le \varepsilon$. This proves the strict confidence claim, including equality in the failure event.

Finally, at fixed L, every reciprocal precision and truncation radius in (2) is polynomially bounded in m, K. Hence $\log q = O_L(m \log(mK))$. Substitution into (3) proves the asserted growth bound. □

The next research goal is a width threshold polynomial in d, C and the accuracy parameters, followed ideally by a polynomial sample guarantee at every $N \ge C$. The present proof does not supply either improvement. Its costly step is a uniform quantization of all coordinates of a Gaussian field on the entire cube. The quadratic-tilt compression used for counting is dimension efficient, but it applies to variational minimizers. A finite-width counterpart that controls the aggregate mass of networks without first quantizing an arbitrary 2^d-dimensional field would be needed to obtain a substantially sharper threshold by this strategy.

For compatibility with the variational notation below, write $\phi = \tanh$ and $\mu_{N,1} = \mu_N$.

A An all-width target-mass bound

The following construction bounds the target's prior cost at every width $N \ge C$, at the mean-field exponent.

Proposition A.1 (Uniform cloning mass bound)**.** *Fix $L \ge 2$ and a bounded 1-Lipschitz activation. There is a function $Q_L(d, C)$ bounded by a polynomial such that for every Boolean target with $C_L(f) \le C$, and every $N \ge C$,*

$$-\frac{2}{N} \log \Pr\{f(x) z_w(x) > 1/2 \text{ for all } x\} \le Q_L(d, C).$$

The prior variances and absence of later-layer biases are exactly those in the main model.

Proof. Choose a width-C teacher, increase its readout by a factor of three, and write its hidden matrices as B^s and readout as v. Thus the teacher has margin strictly greater than $3/2$, hidden entries bounded by C, and readout entries bounded by $3C$. If the original witness has smaller width, first use its actual width in the construction below; all estimates only improve. Partition $\{1, \dots, N\}$ into C groups of sizes k_a differing by at most one. Then

$$1 \le k_a, \qquad \frac{\max k_a}{\min k_a} \le 2, \qquad \frac{N}{k_a} \le 2C.$$

Let $u_a = k_a^{-1/2} \mathbf{1}_{\text{group } a}$ and let P be the orthogonal projection onto their span. The exact clone w_* has first-layer row B^1_b for a neuron in group b, hidden entry B^s_{ba}/k_a from group a to group b, and readout v_b/k_b on group b. Every hidden activation of the exact clone is group-constant. Moreover,

$$\|W^s_*\|_{\text{op}} \le \sqrt{2} C^2 \quad (2 \le s < L), \qquad \sqrt{N} \|W^L_*\|_2 \le 3\sqrt{2} C^2.$$

The first bound follows by restricting to the group-constant subspace: the resulting matrix has entries $\sqrt{k_b/k_a}B^s_{ba}$, and its Frobenius norm is at most $\sqrt{2}C^2$.

Set $t = 1$. The random coordinates $\sqrt{d}W^1_{ij}$, $\sqrt{N}W^s_{ij}$ for $2 \le s < L$, and $\alpha_j = NW^L_j$ are independent centered Gaussians of standard deviation t. Impose the following cylinder event $\mathcal{C}_\rho$, for $0 < \rho \le 1$:

- every $\sqrt{d}W^1_{ij}$ is within ρt of $\sqrt{d}(W^1_*)_{ij}$;
- for every hidden row j and group a, the coordinate $\langle \sqrt{N}W^s_{j,\cdot}, u_a\rangle$ is within ρt of its cloned value $\sqrt{N/k_a}B^s_{ba}$;
- every α_j is within ρt of its cloned value Nv_b/k_b.

All these constraints concern independent scalar Gaussian coordinates. The orthogonal coordinates of each internal row remain independent prior Gaussians. Consequently write

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad E^s P = 0.$$

On the cylinder, $\|R^s\|_{\mathrm{op}} \le \|R^s\|_F \le \rho t\sqrt{C}$. The free matrix is distributed as $(t/\sqrt{N})G^s(I-P)$, with G^s a standard Gaussian matrix, independently of the cylinder coordinates. Let

$$b_L = 2 + \sqrt{2\log(2L)}.$$

The Gaussian operator-norm bound and a union bound imply, conditionally on $\mathcal{C}_\rho$, that all $\|E^s\|_{\mathrm{op}} \le b_L t$ with probability at least $1/2$. This is also valid when $L = 2$, where there are no such matrices.

Let $e_s(x) = N^{-1/2}\|h^s_w(x) - h^s_*(x)\|_2$. The input constraints give $e_1(x) \le \rho(d+1)/\sqrt{d} \le \rho(d+1)$. Since $E^s h^{s-1}_* = 0$, boundedness and Lipschitz continuity give, simultaneously for every input,

$$e_s(x) \le (\sqrt{2}C^2 + \sqrt{C} + b_L)e_{s-1}(x) + \rho\sqrt{C}.$$

Put $A = 2 + 3C^2 + b_L$. Induction yields $e_s(x) \le \rho(d + L + 1)A^{L-2}$ for $s < L$. At the output,

$$|z_w(x) - z_*(x)| \le 3\sqrt{2}C^2 e_{L-1}(x) + \rho.$$

Hence choosing

$$\rho = \left[2 + 9C^2(d + L + 1)A^{L-2}\right]^{-1}$$

ensures output discrepancy at most $1/2$. All networks in the cylinder with the operator-norm event therefore implement f with margin greater than one.

It remains to price the cylinder. Its number of constrained coordinates is

$$m = N\big[d + 2 + (L-2)C\big].$$

The sum of their squared centers is at most NB_0, where one may take

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4.$$

Indeed, a first-layer center has magnitude at most $\sqrt{d}C$, an internal center at most $\sqrt{2C}C$, and a readout center at most $6C^2$. For $Z \sim N(0, t^2)$ and $|Z - a| \le \rho t$,

$$\Pr\{|Z - a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} \exp\{-a^2/t^2 - \rho^2\}.$$

Multiplication and $t = 1$ show

$$-\log \Pr(\mathcal{C}_\rho) \le NB_0 + N[d + 2 + (L-2)C] \log \frac{e\sqrt{2\pi}}{2\rho}.$$

The desired probability is at least $\Pr(\mathcal{C}_\rho)/2$. Since $N \ge 1$, this proves the claim with

$$Q_L(d, C) = 2B_0 + \frac{4[d + 2 + (L-2)C]}{\rho} + 2.$$

Here we used $\log(e\sqrt{2\pi}/(2\rho)) \le 2/\rho$ for $0 < \rho \le 1$. The displayed expression is a polynomial in d, C at every fixed depth. □

B An exact variational principle at the critical exponent

For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation. It provides the variational foundation for the compression argument that follows.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \dots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1, 1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E hh^{\mathsf{T}}\, d\nu(h), \qquad Q_0(a, b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = (\phi((Q^{1/2}u)_a))_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho}\, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^+ is the Moore–Penrose inverse.

Theorem B.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1, \dots, \nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D\big(\nu_s \| \mathsf{M}_{K(\nu_{s-1})}\big) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{7}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z \in O} I(z) \le \liminf_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z \in F} I(z).$$

The infimum in (7) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [1].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N} \sum_{i=1}^{N} (h_i^{s-1}(x^a) h_i^{s-1}(x^b))_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2} V_N. \tag{8}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta\|\Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s\|\Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N}, \dots, \eta_{r,N}, V_N)$.

The recursive map from these variables to $(\nu_1, \dots, \nu_r, z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j} \, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q \, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (8).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta : T_{\#}\eta = \nu} D(\eta\|\rho) = D(\nu\|T_{\#}\rho). \tag{9}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_{\#}\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the

identity in the extended real sense. Apply (9) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v: Q^{1/2} v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (7).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^\mathsf{T} z - \frac{1}{2} t^\mathsf{T} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (7) is attained. Moreover $K(\nu) \preceq m I_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary B.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1, 1\}^m$, set*

$$J(f) = \inf_{z:\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N \to \infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f}) = J(f). \tag{10}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g \in \mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N \to \infty} -\frac{1}{N} \log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (7) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (10).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g \in \mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

The variational principle exhibits the distinction between the critical regime and a frozen kernel model. At $\gamma = 1$, deviations of a hidden feature law have an order-N entropy cost, at the same scale as an order-one output. Thus the optimization is over every hidden feature law, with a separate entropy charge for each layer. It is not in general legitimate to set those laws to their unconditioned values. The following sections obtain a polynomial description of the minimizing feature laws for tanh and use it to compress every Boolean function with bounded full-margin rate.

C Quadratic tilts at a critical full-margin minimizer

We now specialize to $\phi = \tanh$. Its analyticity and its bounded first two derivatives yield additional structure at the minimizers of (7). The following argument uses the exact variational principle above; it does not assume mean-field equations.

Lemma C.1 (Positive definite feature covariances)**.** *If $(\nu_1, \dots, \nu_r, z)$ has finite cost in (7), then every $K(\nu_s)$ is positive definite.*

Proof. For any nonzero $c \in \mathbb{R}^m$, the function

$$H_c(b, v) = \sum_{a=1}^{m} c_a \tanh(b + v \cdot x^a)$$

is real analytic and is not identically zero. To see the latter, choose v_0 such that the numbers $v_0 \cdot x^a$ are all distinct. If H_c were identically zero, evaluating it at $(b, v) = (-t\theta, tv_0)$ and letting $t \to \infty$ would give

$$\sum_a c_a \operatorname{sign}(v_0 \cdot x^a - \theta) = 0$$

for every θ different from these numbers. The difference between two successive such identities is twice one coefficient c_a, so all coefficients would vanish. A nonzero real-analytic function has a Lebesgue-null zero set. The first-layer weights have a positive Gaussian density on the $(d+1)$-dimensional (b, v) space. Consequently

$$\mathsf{M}_{Q_0}\{h : c^{\mathsf{T}} h = 0\} = 0.$$

Finite entropy implies $\nu_1 \ll \mathsf{M}_{Q_0}$, and therefore $c^{\mathsf{T}} K(\nu_1) c > 0$ for every $c \neq 0$.

Inductively, if $Q = K(\nu_{s-1})$ is positive definite, then M_Q has a positive density on $(-1, 1)^m$, since coordinatewise tanh is a smooth bijection from $\mathbb{R}^m$ onto that set. Every proper linear hyperplane is M_Q-null. Absolute continuity $\nu_s \ll \mathsf{M}_Q$ again gives positive definiteness of $K(\nu_s)$. □

Theorem C.2 (Quadratic form of the optimal feature laws)**.** *Let $a > 0$, $f \in \{-1, 1\}^m$, and suppose*

$$J_a(f) = \inf_{z:\ f_b z_b \geq a\ \forall b} I(z) \leq K < \infty.$$

There is a minimizing tuple $(\nu_1, \dots, \nu_r, z)$. Write $Q_s = K(\nu_s)$ and $\beta = Q_r^{-1} z$. There are symmetric matrices $\Gamma_1, \dots, \Gamma_r$ such that

$$\frac{d\nu_1}{d\mathsf{M}_{Q_0}}(h) = \frac{e^{h^{\mathsf{T}}\Gamma_1 h}}{Z_1}, \qquad \frac{d\nu_s}{d\mathsf{M}_{Q_{s-1}}}(h) = \frac{e^{h^{\mathsf{T}}\Gamma_s h}}{Z_s} \quad (2 \leq s \leq r). \tag{11}$$

Moreover,

$$\|\beta\|_1 \leq \frac{2K}{a}, \qquad \Gamma_r = \frac{1}{2}\beta\beta^{\mathsf{T}}. \tag{12}$$

If $g_s = \sum_{i,j} |(\Gamma_s)_{ij}|$, then

$$g_r \leq \frac{2K^2}{a^2}, \qquad g_s \leq 3g_{s+1} + 2g_{s+1}^2 \quad (1 \leq s < r). \tag{13}$$

Thus, at fixed depth, all these entrywise norms are bounded by a polynomial in K and $1/a$, independently of the number $m = 2^d$ of inputs.

Proof. Existence follows from the goodness of I, closedness of the margin constraints, and attainment of the feature-law infimum in Theorem B.1. Every Q_s is positive definite by Lemma C.1.

For fixed feature laws, the output minimizes the strictly convex quadratic $z^\mathsf{T} Q_r^{-1} z/2$ subject to the linear margin constraints. Their feasible set has a strictly feasible point. The elementary quadratic-program optimality conditions give multipliers $\lambda_b \geq 0$ such that

$$\beta_b = f_b \lambda_b, \qquad \lambda_b (f_b z_b - a) = 0.$$

It follows that

$$\frac{1}{2}\beta^\mathsf{T} z = \frac{a}{2}\sum_b \lambda_b = \frac{a}{2}\|\beta\|_1 \leq J_a(f) \leq K,$$

which proves the first bound in (12).

We justify the feature-law variations, including possible zeros of their densities. Fix a layer s, freeze every other variable, and denote its reference measure by μ_s (either M_{Q_0} or $\mathsf{M}_{Q_{s-1}}$). Its contribution to the objective has the form

$$D(\nu_s \| \mu_s) + F_s(K(\nu_s)) + \text{constant},$$

where F_s is differentiable near the current positive definite Q_s. For $s = r$, $F_r(Q) = z^\mathsf{T} Q^{-1} z/2$, so

$$\nabla F_r(Q_r) = -\frac{1}{2}\beta\beta^\mathsf{T}.$$

For $s < r$, put $\xi(h) = (\operatorname{arctanh}(h_b))_{b=1}^m$ and $B_{s+1} = \int \xi(h)\xi(h)^\mathsf{T}\, d\nu_{s+1}(h)$. This moment is finite: by invariance of relative entropy under the coordinatewise tanh bijection, the preactivation law has finite relative entropy with respect to $\mathcal{N}(0, Q_s)$, and the Gaussian exponential moment of a sufficiently small multiple of $\xi^\mathsf{T} Q_s^{-1}\xi$ controls its expectation by the entropy inequality. The Gaussian density ratio, whose Jacobian cancels under tanh, gives, near Q_s,

$$F_s(Q) = \frac{1}{2}\log\det Q + \frac{1}{2}\operatorname{tr}(Q^{-1}B_{s+1}) + \text{constant}.$$

In particular,

$$-\nabla F_s(Q_s) = \frac{1}{2}Q_s^{-1}(B_{s+1} - Q_s)Q_s^{-1} =: \Gamma_s. \tag{14}$$

Put $B = \nabla F_s(Q_s)$ and let ρ_s have density $e^{-h^\mathsf{T} B h}/Z$ with respect to μ_s. Its logarithm is bounded, so $D(\nu_s\|\rho_s)$ is finite. For $\nu_{s,t} = (1-t)\nu_s + t\rho_s$, convexity of relative entropy gives

$$D(\nu_{s,t}\|\rho_s) - D(\nu_s\|\rho_s) \leq -tD(\nu_s\|\rho_s).$$

The identity $D(\nu\|\rho_s) = D(\nu\|\mu_s) + \int h^\mathsf{T} B h\, d\nu + \log Z$ and differentiability of F_s imply that the full objective changes by at most $-tD(\nu_s\|\rho_s) + o(t)$. Minimality forces $D(\nu_s\|\rho_s) = 0$. Thus $\nu_s = \rho_s$, proving (11) with $\Gamma_s = -\nabla F_s(Q_s)$. This also proves $\Gamma_r = \beta\beta^\mathsf{T}/2$.

It remains to bound the apparently ill-conditioned expression (14) without using $\|Q_s^{-1}\|$. Under ν_{s+1} the preactivation vector ξ has Gaussian reference $\mathcal{N}(0, Q_s)$ tilted by $e^{V(\xi)}$, where

$$V(\xi) = \tanh(\xi)^\mathsf{T}\Gamma_{s+1}\tanh(\xi).$$

Twice applying Gaussian integration by parts gives

$$B_{s+1} - Q_s = Q_s\, \mathbb{E}_{\text{tilt}}[\nabla^2 V + (\nabla V)(\nabla V)^\mathsf{T}]\, Q_s.$$

This identity is justified directly because V and its first two derivatives are bounded. Substitution into (14) gives

$$\Gamma_s = \frac{1}{2}\mathbb{E}_{\text{tilt}}[\nabla^2 V + (\nabla V)(\nabla V)^{\mathsf{T}}].$$

The bounds $|\tanh| \le 1$, $|\tanh'| \le 1$, and $|\tanh''| \le 2$ imply

$$\|\nabla V\|_1 \le 2g_{s+1}, \qquad \sum_{i,j} |\partial_i \partial_j V| \le 6g_{s+1}.$$

Therefore $g_s \le 3g_{s+1} + 2g_{s+1}^2$. The final-layer bound is $g_r = \|\beta\|_1^2/2 \le 2K^2/a^2$. Iterating the displayed quadratic recursion a fixed number of times proves the asserted polynomial dependence. □

D Compression of bounded quadratic-tilt towers

This section uses $\phi = \tanh$. Write $\|A\|_{\max} = \max_{ij} |A_{ij}|$ and $\|A\|_{1,\text{ent}} = \sum_{ij} |A_{ij}|$ for matrices on the finite input set. These are different norms; their dual pairing is what avoids a factor of 2^d below. For a positive semidefinite covariance Q and a symmetric matrix Γ, let

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\phi(Z)\phi(Z)^\top \exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}.$$

All activations are coordinatewise. The definition is valid for singular Q.

Lemma D.1 (A dimension-independent covariance comparison)**.** *If $g = \|\Gamma\|_{1,\text{ent}}$, then for all positive semidefinite Q, Q' of the same size,*

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1+g)^2 \|Q - Q'\|_{\max}.$$

Proof. Put $V(z) = \phi(z)^\top \Gamma \phi(z)$ and $Q_t = (1-t)Q + tQ'$. For a twice differentiable bounded test function F, let $\mathbb{E}_t$ denote expectation under the Gaussian of covariance Q_t reweighted by e^V. Gaussian differentiation and the quotient rule give

$$\frac{d}{dt}\mathbb{E}_t F = \frac{1}{2}\sum_{a,b} (Q' - Q)_{ab} \{\mathbb{E}_t[\partial_{ab} F + \partial_a F \partial_b V + \partial_b F \partial_a V] + \operatorname{Cov}_t(F, \partial_{ab} V + \partial_a V \partial_b V)\}.$$

This formula has no reciprocal partition-function bound: all terms are expectations under a probability measure. Define $M_1(H) = \sup_z \sum_a |\partial_a H(z)|$ and $M_2(H) = \sup_z \sum_{a,b} |\partial_{ab} H(z)|$. Because $|\phi|, |\phi'| \le 1$ and $|\phi''| \le 2$,

$$M_1(V) \le 2g, \qquad M_2(V) \le 6g.$$

For $F(z) = \phi(z_i)\phi(z_j)$, including $i = j$, $\|F\|_\infty \le 1$, $M_1(F) \le 2$, and $M_2(F) \le 6$. The absolute derivative is therefore at most

$$\frac{1}{2}\|Q - Q'\|_{\max}\{6 + 8g + 2(6g + 4g^2)\} = (3 + 10g + 4g^2)\|Q - Q'\|_{\max}.$$

Integrating in t proves the assertion, with the displayed slightly larger constant. One may initially replace both covariances by $Q + \epsilon I$ and $Q' + \epsilon I$. The functions and all derivatives used above are bounded, and the partition functions lie between e^{-g} and e^g, so weak Gaussian convergence lets $\epsilon \downarrow 0$. This also proves the singular case. □

Proposition D.2 (Finite realization of a bounded-tilt tower)**.** *Fix $L \geq 2$, $K \geq 1$, and $G \geq 1$. Let Q_0 be the covariance of the Gaussian first-layer field $w \cdot \bar{x}$, with $w \sim \mathcal{N}(0, I_{d+1}/d)$, indexed by $x \in \mathcal{X}$. Suppose symmetric matrices $\Gamma_1, \ldots, \Gamma_{L-1}$ and a vector $\beta \in \mathbb{R}^{\mathcal{X}}$ satisfy*

$$\|\Gamma_s\|_{1,\text{ent}} \leq G, \qquad \|\beta\|_1 \leq 4K, \qquad Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1}).$$

Then the function $z(x) = (Q_{L-1}\beta)_x$ can be uniformly approximated to accuracy $1/8$ by a depth-L network whose width and weights are bounded by $\mathrm{poly}_L(d, K, G)$. If $f(x)z(x) \geq 1/2$ everywhere, a factor-two increase in the finite network's readout implements f with margin at least $3/4$.

Proof. Let $b = 8(1+G)^2$ and

$$\eta = [32K(L-1)b^{L-2}]^{-1}.$$

Choose the common width n to be the least integer satisfying

$$n \geq 2\eta^{-2} \log(8(L-1)\, 2^{2d}).$$

The construction is random only to prove existence. The matrices Γ_s may be indexed by the entire cube; they select the sampling laws, but are not stored or evaluated by the resulting finite network. No efficient sampling or construction algorithm is claimed. At the first layer, independently sample n Gaussian weight vectors $w_i \sim \mathcal{N}(0, I_{d+1}/d)$ tilted by

$$\exp\{h(w_i)^\top \Gamma_1 h(w_i)\}, \qquad h(w_i)_x = \phi(w_i \cdot \bar{x}).$$

Their activation vectors have exactly the law defining $Q_1 = \mathsf{F}_{\Gamma_1}(Q_0)$. At each subsequent hidden layer, conditional on the already sampled previous activation vectors $h_1, \ldots, h_n$, independently sample n standard Gaussian row vectors $\xi_j \in \mathbb{R}^n$ tilted by

$$\exp\{\phi(Z_j)^\top \Gamma_s \phi(Z_j)\}, \qquad Z_j(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \xi_{ji} h_i(x).$$

Use the actual network weights $\xi_{ji}/\sqrt{n}$. If $\widehat{Q}_{s-1} = n^{-1} \sum_i h_i h_i^\top$ is the preceding empirical activation covariance, then, conditionally, Z_j before tilting is Gaussian with exactly that covariance. Thus the new activation vectors are conditionally independent, with second moment $\mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})$.

Every covariance entry is the empirical mean of n variables in $[-1, 1]$. Hoeffding's inequality, a union bound over 2^{2d} entries, and conditioning at successive layers show, with probability at least $3/4$, that

$$\|\widehat{Q}_s - \mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})\|_{\max} \leq \eta \quad (1 \leq s < L),$$

where $\widehat{Q}_0 = Q_0$. Lemma D.1 then implies

$$\|\widehat{Q}_{L-1} - Q_{L-1}\|_{\max} \leq (L-1)b^{L-2}\eta.$$

For each last-layer activation vector h_i, choose the deterministic readout weight $v_i = (\beta \cdot h_i)/n$. The resulting output is $\widehat{z}(x) = (\widehat{Q}_{L-1}\beta)_x$, and therefore

$$\|\widehat{z} - z\|_\infty \leq 4K(L-1)b^{L-2}\eta = 1/8.$$

Although β is indexed by the entire input cube, its only role in the finite network is to select these n real readout weights; each satisfies $|v_i| \leq 4K/n$. It is not an additional computational input to the network.

It remains to bound the sampled weights without losing polynomial size. For every sampled row, its tilted density relative to its Gaussian reference is at most e^{2G}, since the quadratic tilt lies in $[-G, G]$. Let $D_n = n(d+1) + (L-2)n^2$ be the total number of incoming scalar coordinates, and choose

$$B = \left[4G + 2\log(8\max\{D_n, 1\})\right]^{1/2}.$$

Each raw incoming coordinate has reference variance at most one, so, also conditionally on all preceding layers, its probability of exceeding B in absolute value is at most $2e^{2G-B^2/2}$. A union bound gives probability at most $1/4$ that any incoming coordinate exceeds B. The actual internal weights divide the raw coordinates by $\sqrt{n}$ and so satisfy the same bound. Intersecting this event with the covariance-approximation event leaves probability at least $1/2$. Fix a realization in the intersection. Both n and $\max\{B, 8K/n\}$ after the factor-two readout increase are bounded by polynomials in d, K, G at fixed depth. The stated margin follows from $2(1/2 - 1/8) = 3/4$. □

Lemma D.3 (Counting finite realizations)**.** *Fix L, width n, and an entrywise bound $B \geq 1$. The Boolean functions having depth-L, width-n implementations with margin at least $3/4$ and weights bounded by B number at most*

$$\exp\{P\log(3 + 16B\Lambda)\}, \qquad P = n(d+1) + (L-2)n^2 + n, \quad \Lambda = (d + Ln)[1 + n(B+1)]^{L-1}.$$

Proof. Round every weight to a multiple of $(8\Lambda)^{-1}$. The first hidden layer's coordinatewise error is at most $(d+1)/(8\Lambda)$. Every subsequent hidden layer has error at most $n(B+1)$ times the preceding error plus $n/(8\Lambda)$; the same estimate applies at the linear readout. Induction bounds the final error by $1/8$. Each rounded coordinate has at most $3 + 16B\Lambda$ possible values, so the P-parameter grid has the stated size. Rounding preserves every represented Boolean sign function. □

E Critical rate complexity and Bayesian generalization

Theorem E.1 (Polynomial rate-sublevel counting)**.** *For every fixed depth $L \geq 2$, with activation* tanh*, there is a polynomial B_L such that, for every $K \geq 1$,*

$$\log \#\{f : \mathcal{X} \to \{-1, 1\} : J(f) \leq K\} \leq B_L(d, K).$$

Moreover every function in this sublevel set has a depth-L implementation of width and weight magnitude at most $\mathrm{poly}_L(d, K)$*.*

Proof. Take a minimizing tuple for $J(f) \leq K$ in Theorem C.2, with margin $a = 1/2$. Its vector β satisfies $\|\beta\|_1 \leq 4K$. Let $g_r^* = 8K^2$ and recursively $g_s^* = 3g_{s+1}^* + 2(g_{s+1}^*)^2$ for $s < r = L - 1$. Then $G = 1 + \max_s g_s^*$ is a polynomial in K at fixed depth, and all the quadratic tilt matrices have entrywise norm at most G. The feature covariances satisfy $Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1})$, and the optimal output is $z = Q_r\beta$. Proposition D.2 supplies a finite implementation with margin at least $3/4$, common width bounded by a polynomial in d, K, and weights bounded by another such polynomial. The bounds in its construction can be chosen uniformly for every function in the sublevel set. Lemma D.3 counts the resulting fixed parameter grid and gives the stated bound. For the complexity statement, increase width by zero padding if necessary so that the width also bounds every weight magnitude. This preserves the network for tanh, since $\tanh(0) = 0$. □

An explicit count. For completeness, a numerical expression for $B_L(d, U)$, $U \geq 1$, follows directly from the preceding proofs. Put $g_r = 8U^2$, $g_s = 3g_{s+1} + 2g_{s+1}^2$, $G = 1 + \max_s g_s$, and

$$b_* = 8(1+G)^2, \qquad \eta_* = [32Urb_*^{r-1}]^{-1}, \qquad n_* = \left\lceil 2\eta_*^{-2} \log(8r\, 2^{2d}) \right\rceil .$$

Let $D_* = n_*(d+1) + (L-2)n_*^2$ and

$$B_* = \max\{1, \sqrt{4G + 2\log(8\max\{D_*, 1\})}, 8U/n_*\}, \qquad P_* = D_* + n_*.$$

$$\Lambda_* = (d + Ln_*)[1 + n_*(B_* + 1)]^{L-1}.$$

Then one may take

$$B_L(d, U) = P_* \log(3 + 16B_*\Lambda_*).$$

It is polynomially bounded at each fixed depth. It is a count of Boolean functions, not a claim that the optimal tilts can be computed efficiently.

References

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